

Nirma University

Institute of Technology

Semester End Examination (IR/RPR), January - 2025

Master of Computer Application, Semester-I

6CS105CC22: Computer Networks

Roll/

Exam No.

Supervisor's initial
with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. All questions are compulsory. (No Optional Questions)
2. Use a section-wise separate answer book.
3. Figures to the right indicate full marks.
4. Draw neat sketches wherever necessary.
5. Assume suitable data wherever necessary and mention the same.

Section-I

Q-1 : Answer the following questions. [16]

- CO1 (a) Explain the key functions of Session layer and Physical layer of OSI (8)
BL3 (Open systems Interconnection) Reference Model.
CO1 (b) Differentiate between Star, Bus, Mesh and Ring network topologies (8)
BL2 giving suitable diagrams.

Q-2 : Answer the following questions. [18]

- CO1 (a) Discuss various issues of random-access protocols and controlled (6)
BL3 access protocols. Explain with suitable examples.
CO1 (b) Explain Frequency-division multiple access and Time-division multiple (6)
BL2 access methods. Give the disadvantages of each method.
CO2 (c) Why Selective-Repeat ARQ (Automatic Repeat Request) protocol is (6)
BL3 giving better performance as compared to Go-Back-N ARQ? Justify your answer with a proper diagram and critical comments.

Q-3 : Answer the following questions. [16]

- CO2 (a) Assume an 8-bit data-word as 10111011. Using the hamming code (8)
BL3 (non-linear) Error detection and correction method, generate all required check bits and find the corresponding hamming code word for the above-mentioned 8-bit data word. Now, when a single-bit error is introduced at the first MSB bit then how is the error detected using the hamming code method? Discuss it.
CO3 (b) Fill in the entries in the following router table (R1) under the Network (8)
BL3 address column having "?" value. Thereafter, draw the network diagram for the network using the routing table of router R1.

Mask	One of the Address	Network Address	Next-Hop	Address Interface
/27	220.16.76.224	?	-----	M1
/18	145.23.192.0	?	-----	M0
/22	160.26.64.90	?	-----	M3
Default	Default		130.56.12.4	M2

Section - II

Q-4 **Answer the following questions.** [16]

- CO3 (a) Explain the functionalities of Address Resolution Protocol (ARP) and (8)
BL2 Reverse Address Resolution Protocol (RARP) with suitable diagrams. Also, write the difference between them.
- CO3 (b) Justify the need for Congestion detection in the Transport layer with a (8)
BL3 proper diagram. Also, compare the Slow start congestion phase with the congestion avoidance phase.

Q-5 **Answer the following questions:** [16]

- CO3 (a) One of the IPv4 addresses in a block is 169.234.170.82/27. Using this (8)
BL3 data, find the network mask, the number of host addresses in the network, the first host address, and the last address.
- CO3 (b) Explain the TCP data segment and acknowledgment transfer for the (8)
BL2 following scenario using a suitable timeline diagram between two communicating end systems. Also, discuss the error control mechanism used in the following scenarios for the TCP segments.
- i) lost segment
ii) Fast retransmission.

Q-6 **Answer the following questions.** [18]

- CO2 (a) Explain about digital signature. How does it provide security? Explain (8)
BL2 it by using a suitable example.
- CO3 (b) An IP router with a Maximum Transmission Unit (MTU) of 1620 bytes (10)
BL4 has received an IP packet of size 4400 bytes with an IP header of length 20 bytes. To send the IP packet the Router will fragment the packet. Now answer the following:
- i) How many fragments would reach the destination?
ii) For each fragment, find out the value of the More Fragments bits, Fragment Offset, and the Total length of the fragment.